

**SIDO KANHU MURMU UNIVERSITY**  
**DUMKA, JHARKHAND**  
**DEPARTMENT OF GEOGRAPHY**



**FOUR YEARS UNDERGRADUATE PROGRAMME BASED ON NEP-2020**

**SYLLABUS OF GEOGRAPHY**

## PROGRAM LEARNING OUTCOMES

### The programme learning outcomes relating to Honours/Research Degree in Geography:

- \* Demonstrating the understanding of basic concepts in geography.
- \* Demonstrating the coherent and systematic knowledge in the discipline of geography to deal with current issues and their solution.
- \* Display an ability to read and understand maps and topographic sheets to look at the various aspects on the space.
- \* Cultivate ability to evaluate critically the wider chain of network of spatial aspects from global to local level on various time scales as well.
- \* Recognize the skill development in Geographical studies programme as part of career avenues in various fields like teaching, research and administration.
- \* It is also suggested that after the completion of FYUGP Hons./Research, students should be able to demonstrate the knowledge obtained in such way so that they can explore the employability options and service to the society.

### SEMESTER WISE COURSES IN GEOGRAPHY MAJOR-1 FOR FYUGP

#### 2022 onwards

**Table 7: Semester wise Examination Structure in Discipline Courses:**

| Semester   | Courses |                                                 | Examination Structure |                            |                            |                                     |
|------------|---------|-------------------------------------------------|-----------------------|----------------------------|----------------------------|-------------------------------------|
|            | Code    | Papers                                          | Credits               | Mid Semester Theory (F.M.) | End Semester Theory (F.M.) | End Semester Practical/ Viva (F.M.) |
| <b>I</b>   | MJ-1    | EVOLUTION OF GEOGRAPHICAL THOUGHT               | 4                     | 25                         | 75                         | ---                                 |
| <b>II</b>  | MJ-2    | PHYSICAL GEOGRAPHY                              | 4                     | 25                         | 75                         | ---                                 |
|            | MJ-3    | PRACTICAL-I (CARTOGRAPHIC TECHNIQUES)           | 4                     | 25                         | ---                        | 75                                  |
| <b>III</b> | MJ-4    | HUMAN AND SETTLEMENT GEOGRAPHY                  | 4                     | 25                         | 75                         | ---                                 |
|            | MJ-5    | PRACTICAL-II (STATISTICAL METHODS IN GEOGRAPHY) | 4                     | 25                         | ---                        | 75                                  |

|             |                   |                                                         |           |     |     |     |
|-------------|-------------------|---------------------------------------------------------|-----------|-----|-----|-----|
| <b>IV</b>   | MJ-6              | WORLD REGIONAL GEOGRAPHY                                | 4         | 25  | 75  | --- |
|             | MJ-7              | FUNDAMENTALS OF REMOTE SENSING & GIS                    | 4         | 25  | 75  | --- |
|             | MJ-8              | PRACTICAL-III (Map & Graphical representation)          | 4         | 25  | --- | 75  |
| <b>V</b>    | MJ-9              | ECONOMIC GEOGRAPHY                                      | 4         | 25  | 75  | --- |
|             | MJ-10             | GEOGRAPHY OF INDIA & JHARKHAND                          | 4         | 25  | 75  | --- |
|             | MJ-11             | PRACTICAL-IV (INSTRUMENTAL SURVEY)                      | 4         | 25  | --- | 75  |
| <b>VI</b>   | MJ-12             | POPULATION GEOGRAPHY                                    | 4         | 25  | 75  | --- |
|             | MJ-13             | AGRICULTURAL GEOGRAPHY                                  | 4         | 25  | 75  | --- |
|             | MJ-14             | REGIONAL PLANNING AND DEVELOPMENT                       | 4         | 25  | 75  | --- |
|             | MJ-15             | PRACTICAL-V (PHYSICAL SURVEY)                           | 4         | 25  | --- | 75  |
| <b>VII</b>  | MJ-16             | NATURAL RESOURCE MANAGEMENT AND ENVIRONMENTAL GEOGRAPHY | 4         | 25  | 75  | --- |
|             | MJ-17             | SOCIAL AND TRIBAL GEOGRAPHY                             | 4         | 25  | 75  | --- |
|             | MJ-18             | TRANSPORT AND TOURISM GEOGRAPHY                         | 4         | 25  | 75  | --- |
|             | MJ-19             | PRACTICAL-VI (Field Excursion)                          | 4         | 25  | --- | 75  |
| <b>VIII</b> | MJ-20             | Seminar And Presentation                                | 4         | 25  | 75  | --- |
|             | AMJ-1             | URBAN GEOGRAPHY                                         | 4         | 25  | 75  | --- |
|             | AMJ-2             | SOIL AND HYDROLOGY                                      | 4         | 25  | 75  | --- |
|             | AMJ-3             | PRACTICAL-VII (ADVANCE MAJOR PRACTICAL)                 | 4         | 25  | --- | 75  |
|             | <b>OR</b><br>RC-1 | RESEARCH METHODOLOGY                                    | 4         | 25  | 75  | --- |
|             | RC-2              | PROJECT DISSERTATION/ RESEARCH INTERNSHIP/ FIELD WORK   | 8         | --- | --- | 200 |
|             |                   | <b>Total Credit</b>                                     | <b>92</b> |     |     |     |

**: Semester wise Course Code and Credit Points for Minor Courses:**

| Semester   | Common, Introductory, Major, Minor, Vocational & Internship Courses |                                                   | Examination Structure |                            |                            |                                     |
|------------|---------------------------------------------------------------------|---------------------------------------------------|-----------------------|----------------------------|----------------------------|-------------------------------------|
|            | Code                                                                | Papers                                            | Credits               | Mid Semester Theory (F.M.) | End Semester Theory (F.M.) | End Semester Practical/ Viva (F.M.) |
| <b>I</b>   | MN-1A                                                               | INTRODUCTORY GEOGRAPHY                            | 4                     | 15                         | 60                         | 25                                  |
| <b>III</b> | MN-1B                                                               | GEOGRAPHY OF INDIA AND JHARKHAND                  | 4                     | 15                         | 60                         | 25                                  |
| <b>V</b>   | MN-1C                                                               | ENVIRONMENTAL GEOGRAPHY & SUSTAINABLE DEVELOPMENT | 4                     | 15                         | 60                         | 25                                  |
| <b>VII</b> | MN-1D                                                               | CLIMATE CHANGE VULNERABILITY AND ADAPTATION       | 4                     | 15                         | 60                         | 25                                  |
|            |                                                                     | <b>Total Credit</b>                               | <b>16</b>             |                            |                            |                                     |

# SEMESTER I

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## MAJOR COURSE –MJ 1:

### EVOLUTION OF GEOGRAPHICAL THOUGHT

|                                                          |                                 |
|----------------------------------------------------------|---------------------------------|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 | Pass Marks: Th (SIE + ESE) = 40 |
|----------------------------------------------------------|---------------------------------|

(Credits: Theory-04) **60 Hours**

#### Course Objective:

The Learning objective of this course are as follows-

1. To explain the concept, definition and scope of Geography as a distinct discipline
2. To recognize the various branches, streams and school of thought in Geography

#### Course Learning Outcomes:

After the completion of course, the students will have ability to:

- \* Distinguish the paradigms in geography discipline through time
- \* Understand the geographical thinking in different regions of world
- \* Appreciate the past and future trends of world geography in general and Indian geography in particular

#### **Course Content:**

**Unit 1-** Definition, nature and scope of geography, Development of geographical thought in India. Paradigms in Geography

**Unit 2-** Pre-Modern- Early Origins of Geographical Thinking with reference to the Classical and Medieval Philosophies.

**Unit 3-** Modern -Evolution of Geographical Thinking and Disciplinary Trends in Germany, France, Britain, United States of America.

**Unit 4-** Debates - Environmental Determinism and Possibilism, Neo-Determinism/ Probablism Systematic and Regional, Ideographic and Nomothetic.

**Unit 5-** Trends - Quantitative Revolution and its Impact, Behaviouralism, Systems Approach, Radicalism, Feminism; Towards Post-Modernism - Changing Concept of Space in Geography, Future of Geography, Paradigmatic shift in geography

#### **Reference Books:**

- \* Bhat, L.S., (2009): Geography in India (Selected Themes). Pearson
- \* Bonnett, A., (2008): What is Geography? Sage.
- \* Dikshit, R. D., (1997): Geographical Thought: A Contextual History of Ideas, Prentice Hall India.
- \* Freeman, R., (1970): Hundred year of Geography, Hutchinson. London.
- \* Hartshorn, R., (1959): Perspectives of Nature of Geography, Rand MacNally and Co.
- \* Harvey, David., (1969): Explanation in Geography, London: Arnold
- \* Holt-Jensen, A., (2011): Geography: History and Its Concepts: A Students Guide, SAGE.
- \* Hussain, M., (2005): Bhugolik Chintan Ka Itihas, Rawat Publications
- \* Johnston, R. J., (Ed.): Dictionary of Human Geography, Routledge.

- \* Kapur, A., (2001): Indian Geography Voice of Concern, Concept Publications.
- \* Martin Geoffrey J., (2005): All Possible Worlds: A History of Geographical Ideas, Oxford.
- \* Sudepta, Adhikari., (2015): Fundamentals of Geographical Thought, Orient Black Swan Pvt Ltd, Hyderabad

## SEMESTER II

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### MAJOR COURSE- MJ 2: PHYSICAL GEOGRAPHY

|                                                          |  |
|----------------------------------------------------------|--|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 |  |
|----------------------------------------------------------|--|

|                                 |
|---------------------------------|
| Pass Marks: Th (SIE + ESE) = 40 |
|---------------------------------|

(Credits: Theory-04) **60 Hours**

#### Course Objective:

The Learning objective of this course are as follows-

1. To explain the concept, definition and scope of earth systems
2. To recognize the structure of the earth, its atmosphere and describe its characteristic features

#### Course Learning Outcomes:

After the completion of course, the students will have ability to:

- \* To classify earth into various domains according to its physical features
- \* Understand the elements of weather and climate and its impacts at different scales.
- \* Understand the oceanic process and availability of resources.

#### Course Content:

**Unit 1-** Geomorphology: Nature and Scope, Origin of the solar system, Earth: Interior Structure and Isostasy; Earth Movements: Plate Tectonics, Types of Folds and Faults, Earthquakes and Volcanoes.

**Unit 2-** Geomorphic Processes: gradation (erosion and weathering), Cycle of Erosion (Davis and Penck). Evolution of Landforms (Erosional and Depositional): Fluvial, Karst, Aeolian, Glacial, and Coastal;

**Unit 3-** Climatology- Atmospheric Composition and Structure; Insolation, Atmospheric Pressure and Winds

**Unit 4-** Climatic classifications (Koppen's and Thornthwaite) and Regions. Cyclones: Tropical and Temperate Cyclones, Monsoon - Origin and Mechanism, El-Nino.

**Unit 5-** Oceanography- Ocean Floor Topography and Oceanic Water Movements: Waves, Currents and Tides. Ocean Salinity: Distribution and Determinants; Coral Reefs and Marine Deposits.

#### **Reference Books:**

- \* Barry, R. G., and Chorley, R. J., (2009): Atmosphere, Weather and Climate (9th Edition), Routledge, New York.
- \* Critchfield, H. J., (1987): General Climatology, Prentice-Hall of India, New Delhi
- \* Gupta, L.S., (2000): Jalvayu Vigyan(Hindi), Madhyam Karyanvay Nidishalya, Delhi Vishwa Vidhyalaya, Delhi
- \* Lal, D. S., (2006): Jalvayu Vigyan(Hindi), Prayag Pustak Bhavan, Allahabad
- \* Oliver, J. E., and Hidore J. J., (2002): Climatology: An Atmospheric Science, Pearson Education, N. Delhi.
- \* Pinet, P. R., (2008): Invitation to Oceanography (Fifth Edition), Jones and Barlett Publishers, USA,

UK and Canada.

- \* Singh, S., (2009): Jalvayu Vigyan (Hindi),, Prayag Pustak Bhawan, Allahabad
- \* Strahler, A.N., (1987) Modern Physical Geography, John Wiley and Sons, New York, Singapore.
- \* Trewartha, G. T., and Horne L. H., (1980): An Introduction to Climate, McGraw- Hill.

## MAJOR COURSE- MJ 3:

### PRACTICALS-I CARTOGRAPHIC TECHNIQUES

Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100

Pass Marks: Pr (ESE) = 40

(Credits: Practicals-04) 120 Hours

#### Instruction to Question Setter for

#### End Semester Examination (ESE):

There will be one Practical Examination of 3Hrs duration. Evaluation of Practical Examination may be as per the following guidelines:

Experiment = 80 marks

Practical notebook & Viva-voce = 20 marks

#### Course Objective:

The Learning objective of this course are as follows-

1. To explain the concept of scale, cross profiles, and weather maps
2. To familiarise students about topographical maps, various types of projections

#### Course Learning Outcomes:

After the completion of course, the students will have ability to:

- \* Read and prepare maps.
- \* Comprehend locational and spatial aspects of the earth surface.
- \* Use and importance of maps for regional development and decision making.

#### Course Content:

**Unit 1-** Scale-Plain, Comparative, Diagonal

**Unit 2-** Cross Profiles- Serial, Superimposed, Projected, Composite;

**Unit 3-** Topographical Map: Introduction, Interpretation, Identification of physical and cultural features

**Unit 4-** Projection: Simple Conical (one standard and two standard parallel), Bonne's, Mercator's

**Unit 5-** Drawing of Climograph & Hythergraph, Ergograph.

**Practical Record:** Practical record book- at least one exercise from all the topics.

#### **Reference Books:**

- \* Misra, R.P.,(2014): Fundamentals of Cartography (Second Revised and Enlarged Edition), Concept Publishing, New Delhi.
- \* Monkhouse, F. J. and Wilkinson, H. R.,(1973): Maps and Diagrams, Methuen, London.
- \* Robinson, A. H.,(2009): Elements of Cartography (6th Edition), John Wiley and Sons, New York.
- \* Sarkar, A.,(2015):Practical geography: A systematic approach, Orient Black Swan Private Ltd., New Delhi
- \* Sharma, J. P., (2010): Prayogic Bhugol(Hindi), Rastogi Publishers, Meerut.
- \* Singh, R.L. and Singh R.P.B.,(1999): Elements of Practical Geography, Kalyani Publishers, New Delhi.
- \* Singh, R.L. &Dutta, P.K., (2012):Prayogatmak Bhugol(Hindi), Central Book Depot, Allahabad
- \* Singh,R.L.,& Singh, Rana. P.B.,(1991):Prayogtmak Bhugol ke Mool Tatva (Hindi), Kalyani Publishers, New Delhi
- \* Steers, J.A. (1970):An Introduction to the Study of Map Projections, University of London Press,

## SEMESTER III

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### MAJOR COURSE- MJ 4: HUMAN AND SETTLEMENT GEOGRAPHY

|                                                          |                                 |
|----------------------------------------------------------|---------------------------------|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 | Pass Marks: Th (SIE + ESE) = 40 |
|----------------------------------------------------------|---------------------------------|

(Credits: Theory-04) **60 Hours**

#### Course Objective:

The Learning objective of this course are as follows-

1. To explain the concept, definition and themes of human geography
2. To familiarise students about human settlement types and patterns

#### Course Learning Outcomes:

After the completion of course, the students will have ability to:

- \* Know the changing human and cultural landscape at different levels.
- \* Understand patterns and processes of population growth and its implications.
- \* Appreciate the nature and quality of human landscapes

#### Course Content:

**Unit 1-** Introduction: Defining Human Geography; Major Themes; Contemporary Relevance, World migration pattern

**Unit 2-** Space and Society: Cultural Regions; Race; Religion and Language, Racial conflicts

**Unit 3-** Human adaptation to extreme environment- Eskimos, Bushman, Pgymi, Gond

**Unit 4-** Settlements: Types and pattern of Rural Settlements; Classification of Urban Settlements; Trends and Patterns of World Urbanization

**Unit 5-** Primate city, Rank size rule, Central Place theory by Christler,

#### **Reference Books:**

- \* Chandna, R.C., (2017):Population Geography, Kalyani Publishers, New Delhi.
- \* Roy D (2022): Population Geography, 2<sup>nd</sup> Edition, Books & Allied, Kolkata
- \* Daniel, P.A. and Hopkinson, M.F. (1989):The Geography of Settlement, Oliver & Boyd, London.
- \* Hassan, M.I. (2005):Population Geography, Rawat Publications, Jaipur
- \* Hussain, Majid., (2012):Manav Bhugol, Rawat Publications, Jaipur.
- \* Johnston, R., Gregory, D.,& Pratt, G., et al. (2008):The Dictionary of Human Geography, Blackwell Publication.
- \* Jordan-Bychkov., et al., (2006):The Human Mosaic: A Thematic Introduction to Cultural Geography, W. H. Freeman and Company, New York.
- \* Kaushik, S.D., (2010):Manav Bhugol, Rastogi Publication, Meerut.
- \* Maurya, S.D., (2012):Manav Bhugol, Sharda Pustak Bhawan, Allahabad.
- \* Rozenblat., Celine., Pumain., Denise and Velasquez., Elkin Eds. (2018): International and Transnational Perspectives on Urban Systems, Springer, Japan, pages 393.

## MAJOR COURSE- MJ 5:

### PRACTICALS-II- STATISTICAL METHODS IN GEOGRAPHY

|                                                          |                           |
|----------------------------------------------------------|---------------------------|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 | Pass Marks: Pr (ESE) = 40 |
|----------------------------------------------------------|---------------------------|

(Credits: Practicals-04) 120 Hours

#### ***Instruction to Question Setter for***

#### ***End Semester Examination (ESE):***

*There will be one Practical Examination of 3Hrs duration. Evaluation of Practical Examination may be as per the following guidelines:*

*Experiment = 80 marks*

*Practical notebook & Viva-voce = 20 marks*

#### **Course Objective:**

The Learning objective of this course are as follows-

- \* To explain the concept quantitative information in general and Geographical data in particular.
- \* To explain the importance of data analytics. The ways data is collected, or data is taken from different sources.
- \* To familiarise students about methods of graphic data representations

#### **Course Learning Outcomes:**

After the completion of course, the students will have ability to:

- \* Use statistical methods and techniques in geographical analysis
- \* Understand quantitative data, methods of sampling, graphical data representation.

#### **Course Content:**

**Unit 1-** Use of Data in Geography: Significance of Statistical Methods in Geography; Sources of Data, Scales of Measurement (Nominal, Ordinal, Interval and Ratio).

**Unit 2-** Tabulation and Descriptive Statistics: Frequencies (Deciles, Quartiles), Cross Tabulation,

**Unit 3-** Central Tendency (Mean, Median and Mode, Centro-graphic Techniques, Dispersion (Standard Deviation)

**Unit 4-** Sampling: Purposive, Random, Systematic and Stratified. Association and

**Unit 5-** Correlation: Rank Correlation, Product Moment Correlation, and Simple Regression.

***Practical Record:*** Each student will submit a record containing exercises from each topic

#### **\* Reference Books:**

- \* Ajai, S. G. and Sanjaya, S.G. (2009) Statistical Methods for Practice and Research, Sage Publications, New Delhi.
- \* Berry, B. J. L. and Marble, D. F. (eds.): Spatial Analysis A Reader in Geography.
- \* Ebdon, D., (1977): Statistics in Geography: A Practical Approach.
- \* King, L. S., (1969): Statistical Analysis in Geography, Prentice-Hall.
- \* Mahmood, A., 1977: Statistical Methods in Geographical Studies, Concept.
- \* Pal, S. K., (1998): Statistics for Geoscientists, Tata McGraw Hill, New Delhi.
- \* Rogerson, P. A., (2001) Statistical Methods for Geography, Sage Publications, New Delhi.
- \* Sarkar, A. (2013): Quantitative geography: techniques and presentations. Orient Black Swan Private Ltd., New Delhi.

## SEMESTER IV

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### MAJOR COURSE- MJ 6: WORLD REGIONAL GEOGRAPHY

|                                                          |                                 |
|----------------------------------------------------------|---------------------------------|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 | Pass Marks: Th (SIE + ESE) = 40 |
|----------------------------------------------------------|---------------------------------|

(Credits: Theory-04) **60 Hours**

#### **Course Objective:**

The Learning objective of this course are as follows-

1. To explain the physical features, drainage and climatic feature of continents
2. To familiarise students about major physiographic region of continents

#### **Course Learning Outcomes:**

After the completion of course, the students will have ability to:

- \* Locate physical features of the world major continents.
- \* Understand climatic condition and climatic pattern of the continents.
- \* Understand the drainage of the continents

#### **Course Content:**

**Unit 1-** Asia - Physical features, Drainage & Climatic condition & Climatic Regions; agriculture and major industry; geographical account of Great plains of China, Indus basin

**Unit 2-**Europe- Physical features, Drainage & Climatic condition & Climatic Regions, agriculture and major industry, geographical account of Steppe's grassland, Rhine basin

**Unit 3-**North America - Physical features, Drainage & Climatic condition & Climatic Regions, agriculture and major industry, geographical account of Appalachian Highland, Central Plains

**Unit 4-**South America - Physical features, Drainage & Climatic condition & Climatic Regions, agriculture and major industry, geographical account of Pampas, and Amazon Rainforest

**Unit 5-**Australia & New Zealand (Oceania) - Physical features, Drainage & Climatic condition & Climatic Regions, agriculture and major industry, geographical account of Dawns grassland, and Great Sandy desert

#### **References-**

- \* Douglas, L. Johnson.,(2009): World Regional Geography, Tenth edition, Pearson Education Inc, New Jersey.
- \* Baker, A. R. H. and Billinge, M. (forthcoming) Geographies of England: the North-South Divide, Imagined and Real (Cambridge)
- \* Brigham, A. P. 1903 Geographic Influences on American History (Boston)
- \* Brooks, C. E. P. 1926 Climate through the Ages (London).
- \* Hussain, M. (2016) World Geography, Rawat Publications

## **MAJOR COURSE- MJ 7: FUNDAMENTALS OF REMOTE SENSING & GIS**

|                                                                 |                                        |
|-----------------------------------------------------------------|----------------------------------------|
| <b>Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100</b> | <b>Pass Marks: Th (SIE + ESE) = 40</b> |
|-----------------------------------------------------------------|----------------------------------------|

(Credits: Theory-04) **60 Hours**

### **Course Objective:**

The Learning objective of this course are as follows-

- \* To explain the meaning, concept, and definition Remote sensing and GIS, as an important tool in the study and explaining geographic phenomenon
- \* To familiarise students about satellite remote sensing, data processing and interpretation, classification
- \* To aware students about use of GPS and GIS, its principle, working mechanism and applications

### **Course Learning Outcomes:**

After the completion of course, the students will have ability to:

- \* Appreciate the strength and application of remote sensing and GIS
- \* Map the resources, their location and availability
- \* Apply this knowledge for sustainable development

### **Course Content:**

**Unit 1-** Remote Sensing: Meaning, Definition & Scope; Development of Remote Sensing; Components and Process of Remote Sensing; EMR Interaction with Atmosphere and Earth Surface;

**Unit 2-** Remote Sensing Platforms & Sensors, Satellite Imagery Interpretation: Visual & Digital Interpretation Techniques; Elements and Interpretation Keys for Visual Interpretation. (Shape, Size, Colour, Tone, Texture, Association), Image Enhancement Techniques; Application of Remote Sensing

**Unit 3-** Geography & Geographic Information System: Definition & Development of GIS; Elements and components of GIS, Spatial Data: Elements & Types of Spatial Data; Raster & Vector Data Structures;

**Unit 4-** Coordinate Systems, Geo- Referencing of Spatial Data, GIS Database: Creation of Spatial & Non-Spatial Data Base;;

**Unit 5-** Digital Elevation Models (DEM), Basic Principles of Computer Assisted Cartography. Integration of GIS with Remote Sensing & Global Positioning System (GPS)

### **Reference Books:**

- \* Anji Reddy, M. (2008): Textbook of Remote Sensing and Geographic Information System, B.S. Publication, Hyderabad
- \* Campbell, J. B., (2007): Introduction to Remote Sensing, Guildford Press.
- \* Chauniyal, D.D., (2010): Sudur Samvedan evam Bhogolik Suchana Pranali (Hindi), Sharda Pustak Bhawan, Allahabad.
- \* Jensen, J. R., (2004): Introductory Digital Image Processing: A Remote Sensing Perspective, Prentice Hall Inc., New Jersey.
- \* Jensen, J.R. (2007): Remote Sensing of the Environment: An Earth Resource Perspective, Prentice-Hall Inc., New Jersey.

- \* Joseph, G. (2005): Fundamentals of Remote Sensing, United Press India.
- \* Lillisand, T.M., and Kiefer, P.W., (2007): Remote Sensing and Image Interpretation, 6th Edition, John Wiley & Sons, New York.

## MAJOR COURSE- MJ 8: PRACTICALS-III MAP & GRAPHICAL REPRESENTATION

|                                                          |                           |
|----------------------------------------------------------|---------------------------|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 | Pass Marks: Pr (ESE) = 40 |
|----------------------------------------------------------|---------------------------|

(Credits: Practicals-04) **120 Hours**

### ***Instruction to Question Setter for***

### ***End Semester Examination (ESE):***

*There will be one Practical Examination of 3Hrs duration. Evaluation of Practical Examination may be as per the following guidelines:*

*Experiment* = 80 marks

*Practical notebook & Viva-voce* = 20 marks

### **Course Objective:**

The Learning objective of this course are as follows-

- \* To familiarise students about methods of graphic data representations
- \* To familiarise students about geological map and projection.

### **Course Learning Outcomes:**

After the completion of course, the students will have ability to:

- \* Use techniques in geographical analysis
- \* Understand graphical data representation.
- \* Understand and prepare Geological map.
- \* Comprehend locational and spatial aspects of the earth surface.

### **Course Content:**

**Unit 1-** Geological Map and Interpretation.

**Unit 2-** Projection: Cylindrical (Equal area and equi-distant, Gall's Stereographic) Zenithal (Polar Zenithal- Gnomonic and Stereographic Zenithal), Interpreted Sinusoidal and Mollweide projection

**Unit 3;** Diagrammatic Data Presentation –Choropleth, Dot, pie, and Isopleths

**Unit 4 Graphical representation :** Spherical, Proportional Circles, Histogram, Polygons, Ogive

Unit 5 Interpretation of weather maps,

**Practical Record:** Practical record book- at least one exercise from all the topics.

### **\* Reference Books:**

- \* Misra, R.P.,(2014): Fundamentals of Cartography (Second Revised and Enlarged Edition), Concept Publishing, New Delhi.
- \* Monkhouse, F. J. and Wilkinson, H. R.,(1973): Maps and Diagrams, Methuen, London.
- \* Robinson, A. H.,(2009): Elements of Cartography (6th Edition), John Wiley and Sons, New York.

## SEMESTER V

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### MAJOR COURSE- MJ 9: ECONOMIC GEOGRAPHY

|                                                          |                                 |
|----------------------------------------------------------|---------------------------------|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 | Pass Marks: Th (SIE + ESE) = 40 |
|----------------------------------------------------------|---------------------------------|

(Credits: Theory-04) **60 Hours**

#### Course Objective:

The Learning objective of this course are as follows-

- \* To familiarise students about nature, scope and importance of economic geography
- \* To explain the concepts of industrial location, various types of economic activities

#### Course Learning Outcomes:

After the completion of course, the students will have ability to:

- \* Distinguish different types of economic activities and their utilities.
- \* Appreciate the factors responsible for the location and distribution of activities.
- \* Examine the significance and relevance of theories in relation to the location of different economic activities.

#### Course Content:

**Unit 1-** Nature, scope and importance of Economic Geography, Spatial Structure of Economy. Factors Affecting location of Economic Activity with special reference to Agriculture, Industry and Industrial location (Weber's and Losch theory)

**Unit 2-** Primary Economic Activities: Hunting, Fishing, Food gathering, Agriculture and Mining, Subsistence and Commercial Economic Activities; Fishing ground and aquaculture. Issues and Challenges for the Development of fishing and forestry.

**Unit 3-** Secondary Activities: Manufacturing, Concept of Manufacturing Regions (Cotton Textile, Iron and Steel), Special Economic Zones and Technology Parks. Knowledge –based Technologies, Electronic age, spatial information Technology, Telecommunication

**Unit-4** Tertiary Activities: Transport (Land, Air, Water and Pipelines), Trade (National and International) and Services.

**Unit 5-** Economic Growth and Development; Definition, concept of Development and Sustainable Development, Human resource development; concept, Measurement, indicators and component;

#### **Reference Books:**

- \* Alexander, J. W., (1963): Economic Geography, Prentice-Hall Inc., Englewood Cliffs, New Jersey.
- \* Bagchi-Sen, S. and Smith, H. L., (2006): Economic Geography: Past, Present and Future, Taylor

and Francis.

- \* Clark, Gordon L.; Feldman, M.P. and Gertler, M.S., eds. (2000): The New Oxford Handbook of Economic Geography, Oxford Press.
- \* Coe, N. M., Kelly P. F. and Yeung H. W., (2007): Economic Geography: A Contemporary Introduction, Wiley-Blackwell.
- \* Combes, P., Mayer T. and Thisse, J. F., (2008): Economic Geography: The Integration of Regions and Nations, Princeton University Press.
- \* Durand, L., (1961): Economic Geography, Crowell.
- \* Hodder, B. W. and Lee, Roger, (1974): Economic Geography, Taylor and Francis
- \* Knowles, R. & Wareing, J., (2004): Economic and Social Geography Made Simple, Rupa & Co., Kolkata.
- \* Saxena, H.M., (2013): Economic Geography, Rawat Publications, Jaipur.
- \* Siddhartha, K., (2013): Economic Geography, Kishalaya Publications Pvt. Ltd., New Delhi.

## MAJOR COURSE- MJ 10: GEOGRAPHY OF INDIA & JHARKHAND

|                                                          |                                 |
|----------------------------------------------------------|---------------------------------|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 | Pass Marks: Th (SIE + ESE) = 40 |
|----------------------------------------------------------|---------------------------------|

(Credits: Theory-04) **60 Hours**

### Course Objective:

The Learning objective of this course are as follows-

- \* To familiarise students about physical geography of India, its demography, social attributes
- \* To explain the concepts of regionalisation on the basis of physiography, socio-cultural and economic characteristics

### Course Learning Outcomes:

After the completion of course, the students will have ability to:

- \* Understand the physical profile of the country
- \* Study the resource endowment and its spatial distribution and utilization for sustainable development
- \* Synthesis and develop the idea of regional dimensions.

### Course Content:

**Unit 1-** Physical: Location, Physiographic Divisions, Climate: characteristics and classification; Soil and Natural vegetation

**Unit 2-** Population: Distribution and Growth, Structure; Social: Distribution of Population by Race, Caste, Religion, Language, Tribes and their Correlation.

**Unit 3-** Regionalisation of India: Physiographic (R. L. Singh), Socio-Cultural (Sopher), Economic (Sengupta)

**Unit 4-** Economic: Mineral and Power Resources: Distribution and Utilization of Iron Ore, Coal, Petroleum, Gas; Agricultural Production of Rice, Wheat, Cotton and Sugarcane; Industrial Development: Industrial Corridors and Industrial Regions.

**Unit 5-** Regional Account of Jharkhand: Geological structure, Physiography, Drainage, Climate, Natural vegetation, Population and Tribes (Santhal, Oraon, Munda); Economic features: Agriculture, Minerals and Industry -Iron and Steel Industry, Silk, Tourism

### Reference Books:

- \* Deshpande, C. D., (1992): India: A Regional Interpretation, ICSSR, New Delhi.
- \* Douglas, L. Johnson.,(2009): World Regional Geography, Tenth edition, Pearson Education Inc, New Jersey.
- \* Johnson, B. L. C., ed. (2001): Geographical Dictionary of India. Vision Books, New Delhi.
- \* Khullar, D.R. (2014): India: A Comprehensive Geography, Kalyani Publishers, New Delhi.
- \* Majid Husain (2009): Geography of India, Tata McGraw hill Education Private Ltd, New Delhi.
- \* Pathak, C. R. (2003): Spatial Structure and Processes of Development in India. Regional Science Assoc., Kolkata.
- \* Sdyasuk, Galina and P, Sengupta., (1967): Economic Regionalisation of India, Census of India.
- \* Sharma, T.C. (2013): Economic Geography of India. Rawat Publication, Jaipur.
- \* Singh R. L., (1971): India: A Regional Geography, National Geographical Society of India.
- \* Spate O. H. K. and Learmonth A. T. A., (1967): India and Pakistan: A General and Regional Geography, Methuen.
- \* Tirtha, Ranjit (2002): Geography of India, Rawat Pubs., Jaipur & New Delhi.

## MAJOR COURSE- MJ 11: PRACTICALS-IV INSTRUMENTAL SURVEY

Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100

Pass Marks: Pr (ESE) = 40

(Credits: Practicals-04) 120 Hours

### ***Instruction to Question Setter for***

### ***End Semester Examination (ESE):***

*There will be one Practical Examination of 3Hrs duration. Evaluation of Practical Examination may be as per the following guidelines:*

*Instrumental Survey* = 55 marks

*PNB and Viva-voce* = 20 marks

### **Course Objective:**

The Learning objective of this course are as follows-

- \* To familiarise students about various Instruments, methods, tools and techniques of ground survey
- \* To make student learn and apply project development, carrying out primary survey for data collection

### **Course Learning Outcomes:**

After the completion of course, the students will have ability to:

- \* Understand the importance of field work, types of survey and application of instruments for leveling
- \* Handle and apply the instrument to measure height, spot height determination techniques
- \* Synthesis and develop the idea of project work on the basis of secondary and primary survey.

### **Course Content:**

**Unit 1:** Importance of field work, Scope and purpose, Types of survey, Principles and applications of selected survey instruments, Plane Table, Plan preparation, Resection method: two-point problem, three-point problem, all methods.

**Unit 2:** Prismatic Compass: Open and closed traverse, Other smaller instruments: Sextant, Abney Level and Indian Clinometer.

**Unit 3:** Dumpy Level: Traverse Survey, Spot height determination and contour plan preparation,

**Unit 4:** Theodolite: horizontal and vertical (height) measurement, Accessible and inaccessible method. Survey of selected area, Preparation of base map by the use of surveying instruments.

**Reference Books:** Robinson A.H (1995) Elements of Cartography John Wiley & Sons USA

- \* Sarkar A.K.(1997): Practical Geography :A Systematic Approach, Oriental Longman Calcutta
- \* Sharma J.P.(2010): Prayogatmak Bhugol,(Hindi) Sahitya Bhawan, Agra
- \* Monkhouse F.J and Wilkinson HR (1952) Maps and Diagrams, their Compilations and Concentration, Muthuen & Co. London.
- \* Harwel JD, Newson MD. (1973)- Techniques in Physical Geography, Mc. Millan Edu. Ltd. London.
- \* Sarkar, A: Practical Geography – A Systematic Approach.
- \* R.L. Singh (2010) Practical Geography, Sharada Pustak Bhavan, 11, University Road, Allahabad
- \* Kaanetkar and Kulkarni: Surveying and Levelling, Part-I and Part-II.

## SEMESTER VI

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### MAJOR COURSE- MJ 12: POPULATION GEOGRAPHY

|                                                          |                                 |
|----------------------------------------------------------|---------------------------------|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 | Pass Marks: Th (SIE + ESE) = 40 |
|----------------------------------------------------------|---------------------------------|

(Credits: Theory-04) **60 Hours**

#### Course Objective

The Learning objective of this course are as follows-

1. To familiarize student with the nature and scope of Population geography.
2. To make students learn about the population change, and its dynamics

#### Course Learning Outcomes:

After the completion of course, the students will have ability to:

1. Know the pattern of population change and its dynamics.
2. Understand processes of population growth and its implications.
3. Appreciate the growth, distribution and composition of population in different parts of the world

#### Course Content:

**Unit 1:** - Nature and Scope of Population Geography, Population Geography and Demography, Sources of Population Data, Distribution and Density of Population, Distribution and its Pattern in the World, Factors Influencing Distribution of Population in the world.

**Unit 2:-** Concept of Population Composition, Population Change: Growth of Population in the World and India, Components of Population Change, Fertility, Mortality and Migration, Determinants of Fertility and Mortality, Demographic Transition Theory.

**Unit 3:-** Migration - Meaning and Types, Causes and Consequences, Theories of Migration – Ravenstein & Lee.

**Unit 4:-** Population and Resources, Optimum Population, Population Projection, Malthus Population Theory, Population Policy of India

**Unit 5-** Population-Resource Relationship, Population Resource Regions, Trends of world population and policy

#### Reference Books:

- \* Chandna R.C. (2009), Geography of Population, Kalyani Publicishers, Ansari Road, Daryaganj, N. Delhi-2.
- \* Majid Hussain (1999), Human Geography, Rawat Publications, Jaipur.
- \* Trewartha GT. (1959) A Geography of Population, World Patterns, John Wiley and Sons Inc. New York.
- \* Ghosh BN. (1987) Fundamentals of Population Geography, Sterling Publishing Company, New Delhi
- \* R.K. Tripathi ((2000) Populaton Geography, Commonwealth Publishers, New Delhi.
- \* Kayastha, S.L. (1998) Geography of Population, Rawat Publications, Jaipur.
- \* Clerk I (1984) Geography of Population, Approaches and Applications, Pergamon Press, Oxford, UK.
- \* Tiwari, Ram Kumar (2015): Jansankhya Bhugol, Prwalika Publication, Allahabad.

- \* Hiralal (2007): Jansankhya Bugol Ke Mul Tatwa, Radha Publication, New Delhi.
- \* Mourya, S.D. (2011): Jansankhya Bhugol, Sharda Pustak Bhawan, Allahabad.
- \* Dubey, K.K. & Singh, M.B. (2001): Jansankhya Bhugol, Rawat Publication, Jaipur.
- \* Roy, Debjani (2022) Population Geography, Books and Allied publisher, Kolkata

## MAJOR COURSE- MJ 13: AGRICULTURE GEOGRAPHY

Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100

Pass Marks: Th (SIE + ESE) = 40

(Credits: Theory-04) **60 Hours**

### Course Objective

The Learning objective of this course are as follows-

1. To familiarize student about the nature, scope, significance and approaches of agriculture geography.
2. To make students learn about the determinants of agricultural land use, new trends in Indian Agriculture, food security

### Course Learning Outcomes:

After the completion of course, the students will have ability to:

1. Conceptualize the agriculture and its determinants.
2. Get the overview of Indian agriculture regions and systems.
3. Have sound knowledge of agriculture revolutions and food security

### Course Content:

**Unit 1-** Nature and scope, Significance and development of agricultural geography, Approaches to the study of agricultural geography, Origin and dispersal of agriculture, Sources of agricultural data.

**Unit 2-** Determinants of agricultural land use – Physical, economic, social and technological, Land holding and land tenure systems, Land reforms

**Unit 3-** Land use policy and planning, Cropping pattern, Intensity of cropping.

**Unit 4-** Theories of agricultural location based on several multi-dimensional factors, Von Thunen's model and its recent modifications, Whittlesey's classification of agricultural regions, Agro-climatic regions of India.

**Unit 5-** Agriculture in India – Land use and shifting cropping pattern, New trends in Indian agriculture, Green Revolution, White Revolution, Blue Revolution, Problems of Indian agriculture, Agricultural Policy of India, Food security

### References:

- \* Mohammad Shafi (2006): Agricultural Geography, Dorling Kindessley (India) Pv. Ltd. New Delhi.
- \* Negi. B.S. (2003) Indian Agriculture: problems, Progress & Prospects, Vikas publishing house Pvt. Ltd. S. Ansari Road, Daryagani, New Delhi-2.
- \* Majid Hussain (2000): Agricultural Geography, Ed Anmol Publishing Pvt. Ltd. Ansari Road, Daryagani, New Delhi-2.
- \* Shafi M. (1999): Agricultural Geography, Kedarnath Ram Nath, 132, College road, Meetat UP-1.
- \* Singh & Dhillon (2000): Agriculture Geography, Prayag Pustak Bhavan, 20 A, University road, Allahabad-211002, UP.
- \* Jasbir singh (2001): Agriculture geography, Prayog Pustak Bhavan, 20 A, University road, Allahabad-211002, UP.
- \* Memonia CB (1998): Agriculture Problems in India: Prayog Pustak Bhavan, 20 A, University road, Allahabad-211002, UP.
- \* Majid Husain (2007): Systematic Agricultural Geography, Rawat publications, Jawahar Nagar, Jaipur, New Delhi – 92.
- \* Tiwari, R.C., & Singh, B.N. (2015): Krishi Bhugol, Prawalika Publications, Allahabad

## MAJOR COURSE- MJ 14: REGIONAL PLANNING AND DEVELOPMENT

Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100

Pass Marks: Th (SIE + ESE) = 40

(Credits: Theory-04) **60 Hours**

### Course Objective

The Learning objective of this course are as follows-

- \* To familiarize the concept of Region and regional planning, Its need and techniques
- \* To make students learn about the theories and models for regional planning, Indicators of development, Multi-Purpose Projects

### Course Learning Outcomes:

After the completion of course, the students will have ability to:

- \* Identify notable lagging regions and solutions for their overall development
- \* Have comprehensive understanding regarding the different regions and application of different models and theories for integrated regional development.
- \* Select appropriate indicators for the measurement of socio-economic regional development

### Course Content:

**Unit 1-** concept of Region, Types, hierarchy, characteristics and delineation of Regional planning, Geography of regional planning, concept, scope, methods, techniques and need of regional planning. Regionalization of India for Planning (Agro Ecological Zones)

**Unit 2-** Theories and Models for Regional Planning: Growth Pole Model of Perroux; Growth Centre Model in Indian Context; Myrdal, Hirschman, Rostow and Friedmann; evaluation of regional disparities/imbances, method of measuring imbalances

**Unit 3-** Concept of Development, Indicators of development, problems and issues of development, planning process-sectoral, multi-level, decentralized planning,

**Unit 4-** Integrated area development (IADP), planning tribal and hill areas, draught prone areas, command areas in watershed, Border area development, Urban Green belt, Planning for metropolitan region.

**Unit 5-** Niti Ayog, Policy and framework; backward regions: identification and its development- a case study- Dandakaranya, North-East region; Multi-Purpose river valley projects- Damodar, Sardar Sarovar Project.

### Reference Books:

- \* Agyeman, Julian, Robert, D. Bullard and Bob, Evans., (Eds.) (2003): Just Sustainabilities: Development in an Unequal World. London: Earth scan. (Introduction and conclusion.).
- \* Anand, Subhash., (2011): Ecodevelopment : Glocal Perspectives, Research India Press,
- \* Baker, Susan., (2006): Sustainable Development. Milton Park, Abingdon, Oxon; New York, NY: Routledge (Chapter2, "The concept of sustainable development")
- \* Blij, H. J. De., (1971): Geography: Regions and Concepts, John Wiley and Sons.
- \* Friedmann, J. and Alonso W. (1975): Regional Policy - Readings in Theory and Applications, MIT Press, Massachusetts.
- \* Haynes J., (2008): Development Studies, Polity Short Introduction Series.
- \* Misra, R. P., Sundaram, K.V. and V.L.S. Prakasa Rao, (1974): Regional Development planning in India, Vikas Publishing House Delhi.
- \* Peet, R., (1999): Theories of Development, The Guilford Press, New York.
- \* Singh, R.B. (2002): Human Dimensions of Sustainable Development, Rawat Pub., Jaipur, pages
- \* UNDP (2001-04): Human Development Report, Oxford University, Shukla, J (2016)

## MAJOR COURSE- MJ 15: PRACTICALS-V PHYSICAL SURVEY

Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100

Pass Marks: Pr (ESE) = 40

(Credits: Practicals-04) 120 Hours

### ***Instruction to Question Setter for***

### ***End Semester Examination (ESE):***

*There will be one Practical Examination of 3Hrs duration. Evaluation of Practical Examination may be as per the following guidelines:*

*Project Work* = 40 marks

*Practical work* = 15 marks

*Viva-voce* = 20 marks

### **Course Objective**

The Learning objective of this course are as follows-

- \* To develop and expose students to an extensive field survey of wider region of India.
- \* To make students identify various physical landforms, processes, and their impact on human and biological world

### **Learning Outcomes:**

After the completion of course, the students will have ability to:

- \* to conduct an extensive survey of a contiguous wider region of India
- \* identify salient landforms, their genesis and their impact on human life, flora and fauna.
- \* Carrying out extensive field study outside the class room

**Unit 1:** Trace the prominent features of the area to be surveyed. Identify the salient landform features of the selected area on a topographical sheet.

**Unit 2:** Identify the landforms on the surface, while in the field. Also note the agents of erosion, transportation and deposition associated with the landforms.

**Unit 3:** Identify and classify the biodiversity in the area (Flora and Fauna).

**Unit 4:** Observe the relationship of various landforms, flora and fauna with land use, settlement, structure and life style of the people.

### **Note:**

- \* University/College will provide the requisite fund for conducting the survey
- \* Based on observations of the above characteristics, prepare a field survey report. The report need to be supplemented with maps, sketches, diagrams and photographs etc.
- \* The practical exercises should aim at identification of micro-geomorphic features on the ground and their relationship to land use/settlement pattern. This is also a training in Report Writing.
- \* Two written questions in the practical examination based on the physical survey report-
  - \* writing method
  - \* physical survey

### **References-**

1. Creswell, J., (1994): Research Design: Qualitative and Quantitative Approaches, Sage Publications, California.
2. Dikshit, R. D. (2003). The Art and Science of Geography: Integrated Readings, Prentice-Hall of India, New Delhi.
3. Dash and Roy, (2022) Field Work In Social Work Education, Atlantic publisher.

## SEMESTER VII

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### MAJOR COURSE- MJ 16: NATURAL RESOURCE MANAGEMENT AND ENVIRONMENTAL GEOGRAPHY

|                                                          |                                 |
|----------------------------------------------------------|---------------------------------|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 | Pass Marks: Th (SIE + ESE) = 40 |
|----------------------------------------------------------|---------------------------------|

(Credits: Theory-04) 60 Hours

#### Course Objective:

The Learning objective of this course are as follows-

- \* To familiarize the concept of Natural resource management, and its concepts
- \* To make students learn about the Ecosystem, its structure, functions and various policy with regard to environmental conservation

#### Course Learning Outcomes:

After the completion of course, the students will have ability to:

- \* Understand the dynamic interactive relationship between man and environment.
- \* Have sound understanding on distribution, utilization and proper management of natural resources at global level.
- \* Make assessment and review of planning and policies related to environment and natural resources.

#### Course Content:

**Unit 1-** Environment and Natural Resource Management: Concept, Human-Environment Relationships;

**Unit 2-** Ecosystem: Concept, Structure and Functions.

Environmental Issues in Tropical, Temperate and Polar Ecosystems.

**Unit 3-** Natural Resource: Concept, Classification; Distribution, Utilization, Problems and Management of Land, Water Forests and Energy.

**Unit 4-** Appraisal and Conservation of Environment and Natural Resources and Sustainable Resource Development.

**Unit 5-** Environmental Programmes and Policies – Global, National and Local levels

#### Reference Books:

- \* Chandna, R. C., (2002): Environmental Geography, Kalyani, Ludhiana.
- \* Cunningham, W. P. and Cunningham, M. A., (2004): Principals of Environmental Science: Inquiry and Applications, Tata Macgraw Hill, New Delhi.
- \* Goudie, A., (2001): The Nature of the Environment, Blackwell, Oxford.
- \* Holechek, J. L. C., Richard, A., Fisher, J. T. and Valdez, R., (2003): Natural Resources: Ecology, Economics and Policy, Prentice Hall, New Jersey.
- \* Miller, G. T., (2004): Environmental Science: Working with the Earth, Thomson Brooks Cole, Singapore.
- \* Mitchell, B., (1997): Resource and Environmental Management, Longman Harlow, England.
- \* MoEF, (2006): National Environmental Policy-2006, Ministry of Environment and Forests, Government of India.
- \* Odum, E. P. et al, (2005): Fundamentals of Ecology, Ceneage Learning India.

- \* Saxena, H.M., 2012: Environmental Studies, Rawat Publications, Jaipur.
- \* Singh, Savindra., (2001): Paryavaran Bhugol (Hindi), Prayag Pustak Bhawan, Allahabad. (in Hindi)
- \* UNEP, (2007): Global Environment Outlook: GEO4: Environment for Development, United Nations Environment Programme.

## MAJOR COURSE- MJ 17: SOCIAL AND TRIBAL GEOGRAPHY

|                                                          |                                 |
|----------------------------------------------------------|---------------------------------|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 | Pass Marks: Th (SIE + ESE) = 40 |
|----------------------------------------------------------|---------------------------------|

(Credits: Theory-04) **60 Hours**

### **Course Objective:**

The Learning objective of this course are as follows-

- \* To familiarize the about social geography, its concept, nature and scope; migration social categories
- \* To make students learn about Tribal geography and its concepts; Tribes and their economic activities, marriage, faith and practices

### **Learning Outcomes:**

After the completion of course, the students will have ability to:-

- \* Understand the nature, scope and relationships of geography and human wellbeing;
- \* Acquire knowledge on spatial dimensions of social diversity components;
- \* Understand the aspects of Tribal geography and tribal socio-economic activities

### **Course Content:**

**Unit 1-** Social Geography: Concept, Origin, Nature and Scope.

**Unit 2-** Peopling Process of India: Technology and Occupational Change; Migration.

**Unit 3-** Social Categories: Caste, Class, Religion, Race and Gender and their Spatial distribution

**Unit 4-** Tribal Geography- meaning, concept, and scope of tribal geography; Tribes and their habitat- Geographical distribution of Indian tribes, groups and sub-groups; Economic activities; Socio- Political Organization- Family, Marriage and kinship, faith, beliefs and practices,

**Unit 5-** Tribal rights- Land, forests, water; Emerging social problems- Health and education, malnutrition, illiteracy, Alcoholism; Industrialization and tribe, mining and tribes, displacement .

### **References :**

- \* Ahmed A., 1999: Social Geography, Rawat Publications.
- \* Casino V. J. D., Jr., 2009) Social Geography: A Critical Introduction, Wiley Blackwell.
- \* Cater J. and Jones T., 2000: Social Geography: An Introduction to Contemporary Issues, Hodder Arnold.
- \* Panelli R., 2004: Social Geographies: From Difference to Action, Sage.
- \* Rachel P., Burke M., Fuller D., Gough J., Macfarlane R. and Mowl G., 2001: Introducing Social Geographies, Oxford University Press.
- \* Smith D. M., 1977: Human geography: A Welfare Approach, Edward Arnold, London.

- \* Smith S. J., Pain R., Marston S. A., Jones J. P., 2009: The SAGE Handbook of Social Geographies, Sage Publications.
- \* Sopher, David (1980): An Exploration of India, Cornell University Press, Ithaca
- \* Valentine G., 2001: Social Geographies: Space and Society, Prentice Hall.

## MAJOR COURSE- MJ 18: TRANSPORT AND TOURISM GEOGRAPHY

Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100

Pass Marks: Th (SIE + ESE) = 40

(Credits: Theory-04) **60 Hours**

### Course Objective:

The Learning objective of this course are as follows-

- \* To be aware of the various dimensions of Tourism Geography and make the students aware about various types of tourism
- \* To assess sustainable ecotourism and other contemporary forms of tourism
- \* To critically evaluate the infrastructure in tourism in India along with reviewing the tourism policy

### Course Learning Outcomes:

After the completion of course, the students will have ability to:

- \* Learn the concept of transport, its types and factors
- \* Equip with a basic understanding of nature and scope, trends and patterns of various types of tourisms
- \* Apply the principles of Geo-tourism and analyse the prospects and problems associated with pilgrimage tourism.

**Unit- I** Nature, scope, significance of and Development of Transport Geography, factors of Development: Physical, Economic, Social, Economic and transport and regional Modes development, relative significance of transport (railways, roadways, waterways ).

**Unit- II** Accessibility and flow models; network structure, measurement of accessibility, Models of network change, Function, pattern of movement and transport development.

**Unit- III** Nature, scope and extent, concept of tourism, Relationship between geography and Tourism, Eco- tourism, Geotourism, Agro-tourism, Heritage Religions tourism and Adventure tourism.

**Unit-IV** Types of tourism- Domestic and the international, Adventure, wildlife, Pilgrimage, Business, Leisure, Pleasure, and cultural tourism, Local, National and international, Socio-Economic impact of tourism.

**Unit-V** Infrastructural approach for the development of tourism, Govt. policies for Planning and Promotion of tourism in India, prospect and manning of tourism in India. Case studies: Hill Station – Mount Abu, Shimla, Ooty, Beach points- Kwalum, Goa and Mariano Beach, Historical Centre – Mysore, Jaipur, Delhi, Religious- Puri, Deoghar Tirupati, Kedarnath, Mahakal (Ujjain); Dams- Tehri, Hirakud, Masanjor National Parks - Palamu Tiger reserve, Kanjiranga and Gir.

### **References-**

- \* Hagget, F and Chorley, R.J. Network analysis, Edward Arnold, London. 1973
- \* Raza, M and Agrawal, Y.P., Transport Geography in India. Concept Publication New Delhi, 1985.
- \* White, H.P. and Senior, M.L; Transport- Longmon London, 1983.

- \* Ulman, E.L. American Commodity flow, University of Washington press, 1957.
- \* Bhatia, A.K. (1996) Tourism Development sterling Publisher, New Delhi.
- \* Singh, R.L. and Kashi Nath Singh; Reding in Rural Settlement, Geographers.
- \* Sharma , J.K. (2000) Tourism, Plannings, and Development – A New perspective Kanishks.

## MAJOR COURSE- MJ 19: PRACTICALS-VI - FIELD EXCURSION

|                                                          |                           |
|----------------------------------------------------------|---------------------------|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 | Pass Marks: Pr (ESE) = 40 |
|----------------------------------------------------------|---------------------------|

(Credits: Practicals-04) **120 Hours**

### End Semester Examination (ESE):

There will be one Practical Examination of 3Hrs duration. Evaluation of Practical Examination may be as per the following guidelines:

Tour report

= 55 marks

Viva-voce

= 20 marks

### Course Objective:

The Learning objective of this course are as follows-

- \* To develop and expose students to an extensive field survey of wider region of India.
- \* To make students identify various physical landforms, processes, and their impact on human and biological world

### Learning Outcomes:

After the completion of course, the students will have ability to:

- \* to conduct an extensive survey of a contiguous wider region of India
- \* identify salient landforms, their genesis and their impact on human life,
- \* Carrying out extensive field study outside the class room

### Course Content:

#### **Field Excursion : Preparation of tour report**

- \* Based on observations prepare a tour report. The report need to be supplemented with maps, sketches, diagrams and photographs etc.
- \* The report should aim at identification of different geographical dimension.
- \* Two written questions in the practical examination based on concept and significance of field excursion

Case study based on field survey should be prepared on a specific theme. Prior approval of Department/ College should be taken

### References:

1. Monkhouse, F. J. and Wilkinson, H. R.,(1973): *Maps and Diagrams*, Methuen, London.
2. Cuff, J. D. and Mattson, M. T., (1982): *Thematic Maps: Their Design and Production*, Methuen Young Books
3. Dent, B. D., Torguson, J. S., and Holder, T. W., (2008): *Cartography: Thematic Map Design* (6th Edition), McGraw Hill Higher Education
5. Kraak, M.J. and Ormeling, F., (2003): *Cartography: Visualization of Geo-Spatial Data*, Prentice-Hall.

## SEMESTER VIII

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### MAJOR COURSE- MJ 20:

#### PAPER: SEMINAR AND GROUP DISCUSSION

FULL MARKS: 100

#### Course Objectives:

To enhance students' communication skills through active participation in seminars and group discussions.

To prepare students for professional and academic settings where effective communication and collaboration are essential.

#### Course Outcomes:

Students will be prepared to apply the communication and collaboration skills developed in this course to professional and academic contexts, enhancing their overall employability and academic success.

#### Course Contents:

The Head of the Department will divide all the students of the class into different groups and each group will be allotted topics related to discipline of geography. Each group will be supervised by a teacher. Each student will prepare individual report.

#### Steps of Evaluation:

1. Group discussion will be held within each group under the chairmanship of Head of the Department at the department level.
2. Presentation and the report submitted by the students will be evaluated by one external member and one internal member.

#### Evaluation Process:

Evaluation of the dissertation work may be as per the following guidelines:

Report= 50 marks

Presentation = 25marks

GROUP DISCUSSION = 25 marks

## ADVANCE MAJOR COURSE- AMJ 1: URBAN GEOGRAPHY

|                                                          |                                 |
|----------------------------------------------------------|---------------------------------|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 | Pass Marks: Th (SIE + ESE) = 40 |
|----------------------------------------------------------|---------------------------------|

(Credits: Theory-04) **60 Hours**

### Course Objective:

The Learning objective of this course are as follows-

- \* To familiarise students about Urbanization, its patterns and theories
- \* To make student learn about urban functions, urban sprawl, urban renewal-policies, Master plan

### Course Learning Outcome:

After the completion of course, the students will have ability to:

- \* Understand the fundamentals and patterns of urbanization process
- \* Learn the functional classification of cities and Central Place Theory
- \* Know about policies of urban development of India

### Course Content:

**Unit 1-** Nature and scope of Urban Geography-Definition of Urban Settlements (Towns, Cities and Metro etc.), Attributes of urban places during ancient, medieval and modern period,

**Unit 2-** Classification of urban settlements on the basis of size and function, Urban growth and theories, Central Place theory of Christaller and Losch, Contribution of Indian scholars to the studies of urban settlements.

**Unit 3-** Urban Population Density and Land Value Curves- Urban Land Use – Vertical and Horizontal Growth of Cities, Concentric, Zonal and Multiple Nuclei Theories of Urban Structure.

**Unit 4-** Urban Functions- Basic and Non-Basic- Urban Hierarchy- Rank-Size Rule – Central Place Theory – Functional Classification of Towns by C.D. Harris and H.J. Nelson. Urban Issues & Challenges: Water supply, traffic congestion, solid waste, smog, sewage and drainage system; Slum and housing problems

**Unit 5-** Concept of City, Region and Urban Hinterland – Urban Sprawl- Urban Slums – Urban Crimes and their Trends with reference to India- Concept and Issues of Peri-Urbanization. Elements of Urban Planning – Urban Renewal – Policies of Urban Development in India – Master Plans of Ranchi City.

### References:

- \* Bansal, S.C. (2011): Nagariya Bhogol. Meenakshi Publication, Meeruth.
- \* Beanjen-Garnier J&G. Chabot (1967) Urban Geography, Jhonwiley, New York.
- \* Johnson James H (1966) Urban Geography – An Introductory Analysis, Pergamon Press Oxford, London.
- \* Karen Stromme Christensen (1999) Cities and Complexity, University of California, Berkely USA, Sage Publication, New Delhi.
- \* Mandal R.B. (2002) Urban Geography – A Text Book, Concept Publishing Company, New Delhi.
- \* Mayer H.M. & Kohn CF (1967) Urban Geography, Central Depot, Allahabad, India
- \* Northham Ray M. (1975) Urban Geography, Jhon Wiley & Sons, Inc. New York
- \* Peter Roberts (2000) Urban Regeneration, University of Dundee, U.K., Sage Publication, New

Delhi.

- \* Ranan Paddison (2001) Hand Book or Urban Studies, University of Glasgow, U.K., Sage Publications, N. Delhi.
  - \* Saskia Sassen (2000) Cities in a World Economy, University of Chicago, USA, Sage Publications, New Delhi.
  - \* Siddartha K & S. Mukherjee (1996). Cities, Urbanization and Urban Systems, Transworld Media and Communication Pvt. Ltd. New Delhi
  - \* Stephen Ward (2004) Planning and Urban Change, Sage Publications, New Delhi
-

## ADVANCE MAJOR COURSE- AMJ 2: SOIL AND HYDROLOGY

|                                                          |  |
|----------------------------------------------------------|--|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 |  |
|----------------------------------------------------------|--|

|                                 |
|---------------------------------|
| Pass Marks: Th (SIE + ESE) = 40 |
|---------------------------------|

(Credits: Theory-04) **60 Hours**

### Course Objective:

The Learning objective of this course are as follows-

- \* To familiarise students about nature, scope and significance of soil geography
- \* To make student learn about soil development, hydrology, hydrological cycle, surface and ground water and its management

### Course Learning Outcome:

After the completion of course, the students will have ability to:

- \* Study the soil as a basic resource, focusing its distribution, problems and management.
- \* Understand the basic components of hydrological cycle and comprehend practices of integrated watershed management.
- \* Evaluate the water balancing and river basin and water disputes

### **Course Contents-**

**Unit 1-** Nature, scope and significance of Soil Geography; its relationship with Pedology, Soil forming factors: parent material, organic, climatic, topographic, Spatio-temporal dimensions, Processes of soil formation and soil development: Physical, Biotic and Chemical. Soil profile.

**Unit 2-** Soil organism, macro-animals (earthworms, sowbugs, mites, centipedes, rodents and insects), Micro-animals and plants-Nematodes, Protozoa, Rotifers, Fungi, Bacteria, algae and Actinomyces

**Unit 3-** Physical properties of soils: Morphology, Texture, Structure, Water, Air, Temperature and other properties of soil, Chemical properties of soil and soil reaction, Soil erosion, Degradation and Conservation

**Unit 4-** Definition and scope of hydrology, importance of water, hydrological cycle, water storages – glaciers, river channels, lakes and reservoirs, soil moisture;

**Unit 5-** Ground water: characteristics of stream flow, Darcy's law, permeability, infiltration, ground water storage, ground water aquifers in different rock systems, movement and discharge. Water Crisis: a Case study- rural or urban, water management: ground water and surface water

### **References:**

- \* Miller, R. W. and Donahue, R. L. (1992): Soils: An Introduction to Soils and Plant Growth, Prentice-Hall of India, New Delhi
- \* Brady, N. C., and Weil, R. R. (2008): The Nature and Properties of Soils, Prentice Hall, New Jersey
- \* Pitty, A. F. (1978): Geography and Soil Properties, Methuen and Co., London
- \* Bridges, E. M. and Davidson, D. A. (1982): Principles and Applications of Soil Geography, Longman Group, London
- \* Birkeland, P. W (1999): Soils and Geomorphology, Oxford University Press, New York
- \* Govinda Rajan, S.V. and Gopala Rao, H.G.: Studies on soils of India, Vikas, New Delhi, 1978.
- \* Raychoudhuri, S.P.: Soils of India, ICAR, New Delhi, 1958.
- \* Bunting, B.T.: The Geography of Soils, McGraw Hill, New York.

- \* Timothy, Davie. 2003. Fundamentals of Hydrology. Routledge, Taylor and Francis Group, U.K.
  - \* Todd, D.K. 2009. Groundwater Hydrology. John Wiley & Sons Inc.
  - \* Mahajan, G. 1989. Evaluation and Development of Groundwater. Ashish Publishing House, New Delhi.
  - \* Karanth, K.R.C. 1988. Ground Water: Exploration, Assessment and Development. Tata-Mcgraw Hill, New Delhi.
  - \* Andrew D. Ward and Stanley Trimble. 2004(2nd edition). Environmental Hydrology. Lewis Publishers.
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## ADVANCED MAJOR COURSE- AMJ 3: PRACTICALS-I ADVANCE MAJOR PRACTICAL

|                                                          |                           |
|----------------------------------------------------------|---------------------------|
| Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 | Pass Marks: Pr (ESE) = 40 |
|----------------------------------------------------------|---------------------------|

(Credits: Practicals-04) 120 Hours

### *Instruction to Question Setter for*

#### End Semester Examination (ESE):

There will be one Practical Examination of 3Hrs duration. Evaluation of Practical Examination may be as per the following guidelines:

Experiment = 80 marks

Practical notebook & Viva-voce = 20 marks

### **Course Objective:**

The Learning objective of this course are as follows-

- \* To familiarise students about drainage density, drainage texture and stream ordering
- \* To make student learn and apply methods of soil testing, planning of satellite and garden town

### **Course Learning Outcome:**

After the completion of course, the students will have ability to:

- \* Estimate the soil quality such as soil pH, macro nutrients, identification of soil problems and management.
- \* Understand stream ordering techniques, calculation of bifurcation ratio.
- \* Evaluate the traffic flow through diagrams, water budget, rainfall dispersion

### **Course Content:**

**Unit 1-** Stream Ordering, Bifurcation ratio, Drainage Density, Drainage Texture, Thalweg, Channel Profiles, Hypsometric Curve, Profiles,

**Unit 2-** Study of Soil P<sub>H</sub> Value, Nitrogen Content, Phosphorous and Construction of Soil Profiles.

**Unit 3-** Traffic Flow Diagram. Water Budget, Rainfall Dispersion Diagram

**Unit 4-** Identification of rocks & minerals, characteristic & classification of minerals.

**Practical Record-** Exercise on each topic above

### **References:**

1. Andrew. D. ward, and Stanley, Trimble., (2004): *Environmental Hydrology*, 2nd edition, Lewis Publishers, CRC Press.
2. Fetter, C.W. (2005): *Applied Hydrogeology*, CBS Publishers & Distributors, New Delhi.
3. Reddy, K. Ramamohan, Venkateswara Rao, B, Sarala, C., (2014): *Hydrology and Watershed Management*, Allied Publishers.
4. Karanth, K.R., (1988): *Ground Water: Exploration, Assessment and Development*, Tata- McGraw Hill, New Delhi.
5. Lyon, J.G., (2003): *GIS for Water Resource and Watershed Management*, Taylor and Francis, New York.

## **ADVANCED MAJOR COURSE- AMJ 3: Research Course (RC 1)**

### **Research Methodology**

|                                                                 |  |                                  |
|-----------------------------------------------------------------|--|----------------------------------|
| <b>Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100</b> |  | <b>Pass Marks: Pr (ESE) = 40</b> |
|-----------------------------------------------------------------|--|----------------------------------|

(Credits: Practicals-04) **120 Hours**

#### **Course Outcomes**

1. This course will help them to select an appropriate research design.
2. With the help of this course, students will be able to take up and implement a research project/ study.
3. The course will also enable them to collect the data, edit it properly and analyse it accordingly. Thus, it will facilitate students' prosperity in higher education.
4. The Students will develop skills in qualitative and quantitative data analysis and presentation.

#### **Module-I**

Meaning, Objective, Significance and motivation of research; Research methods vs Research methodology; Method of geographical studies, Types of research.

#### **Module-II**

Research Design, Research Problems. Hypothesis, theories, law and models

#### **Module- III**

Data collection, Data Processing and Data Analysis, Data interpretation, Sampling.

#### **Module - IV**

Report Writing, Criteria of good research, Problems encountered by researchers in India, Recent Trends in Research

## MINOR COURSE-1A

(SEM-I)

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### MINOR COURSE- MN 1A: INTRODUCTORY GEOGRAPHY

|                                                         |  |
|---------------------------------------------------------|--|
| Marks: 15 (5 Attd. + 10 SIE: 1Hr) + 60 (ESE: 3Hrs) = 75 |  |
|---------------------------------------------------------|--|

|                                 |
|---------------------------------|
| Pass Marks: Th (SIE + ESE) = 30 |
|---------------------------------|

(Credits: Theory-03) **45 Hours**

#### Course Objective:

The Learning objective of this course are as follows-

- \* To familiarise students about the Earth system, its origin, interior
- \* To make student learn about evolution of landforms, structure and composition of atmosphere,
- \* To aware about population distribution, human races, religion and languages

#### **Learning Outcomes:**

After the completion of the course, the students will have the ability to:

- \* Appreciate the fundamental concepts of geography;
- \* apply day to day issues in society and its relation with geography
- \* to understand population growth, social diversity

#### Course Content:

**Unit 1-** Origin of the Earth, Interior structure of the earth, Earthquake and volcanoes

**Unit 2-** Evolution of landforms- Fluvial, glacial, Aeolian, coastal

**Unit 3-** Structure and composition of atmosphere, pressure belt and planetary winds and climatic regions

**Unit 4-** Distribution of human races, religion, language

**Unit 5-** Distribution, density, and growth of World population

#### **Reference Books;**

- \* Bloom, A. L., (2003): Geomorphology: A Systematic Analysis of Late Cenozoic Landforms, Prentice-Hall of India, New Delhi.
- \* Singh Savindra (2015): Bhuakriti vigyan ka Swarup, Prayag Pustak Bhawan, Allahabad
- \* Christopherson, R. W. and Birkeland, G. H., (2012) Geosystems: An Introduction to Physical Geography (8th edition), Pearson Education, New Jersey.
- \* Das Gupta, A and Kapoor, A.N., (2001) Principles of Physical Geography, S.C. Chand & Company Ltd. New Delhi.
- \* Chandna, R.C., (2017):Population Geography, Kalyani Publishers, New Delhi.

- \* Roy D (2022): Population Geography, 2nd Edition, Books & Allied, Kolkata
- \* Hassan, M.I. (2005): Population Geography, Rawat Publications, Jaipur
- \* Hussain, Majid., (2012): Manav Bhugol, Rawat Publications, Jaipur.

## **MINOR COURSE- MN 1A PR:**

### **MINOR PRACTICALS-1A PR- TOOLS AND TECHNIQUES OF GEOGRAPHY**

|                                   |                                  |
|-----------------------------------|----------------------------------|
| <b>Marks: Pr (ESE: 3Hrs) = 25</b> | <b>Pass Marks: Pr (ESE) = 10</b> |
|-----------------------------------|----------------------------------|

(Credits: Practicals-01) **30 Hours**

#### ***Instruction to Question Setter for***

#### **End Semester Examination (ESE):**

There will be one Practical Examination of 3Hrs duration. Evaluation of Practical Examination may be as per the following guidelines:

|                           |            |
|---------------------------|------------|
| Experiment                | = 15 marks |
| Practical record notebook | = 05 marks |
| Viva-voce                 | = 05 marks |

#### **Course Objective:**

The Learning objective of this course are as follows-

- \* To familiarise students about the concept of scale, RF
- \* To make student learn about topographical maps, conventional sign and interpretation

#### **Learning Outcomes:**

After the completion of the course, the students will have the ability to:

- \* Appreciate the concepts of scale, RF;
- \* apply information from topographical maps
- \* to understand conventional signs and symbols

#### **Course Content:**

**Unit 1-** Scale- simple linear scale and RF

**Unit 2-** Study of Topographical Maps- Conventional signs and Interpretation (one each-hilly/plain area)

#### **Practical Record:**

A Project File comprising at least one exercise each, on scale and interpretation of topographic sheet

#### **Reference Books;**

- \* Anson, R., and Ormelling F. J., (1994): International Cartographic Association: Basic

Cartographic, Vol. Pregmen Press.

- \* Singh, Gopal., (1998): Map Work and Practical Geography (4th Edition), Vikas Publishing House, Ahmedabad.
- \* Gupta, K.K. and Tyagi V.C., (1992): Working with Map, Survey of India, DST, New Delhi.
- \* Kraak, M.J., (2010): Cartography: Visualization of Geospatial Data (3rd edition), Pearson Education Ltd., London.

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## **MINOR COURSE - MN1B (SEM-III)**

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### **GEOGRAPHY OF INDIA AND JHARKHAND**

|                                                                |                                        |
|----------------------------------------------------------------|----------------------------------------|
| <b>Marks: 15 (5 Attd. + 10 SIE: 1Hr) + 60 (ESE: 3Hrs) = 75</b> | <b>Pass Marks: Th (SIE + ESE) = 30</b> |
|----------------------------------------------------------------|----------------------------------------|

(Credits: Theory-03) **45 Hours**

#### **Course Objective:**

The Learning objective of this course are as follows-

- \* To familiarise students about the physical features, climate and vegetation of India and Jharkhand
- \* To make student learn about economic, and agricultural features of India and Jharkhand

#### **Learning Outcomes:**

After the completion of course, the students will have ability to:

- \* Understand the physical profile of the India and Jharkhand
- \* Study the resource endowment and its spatial distribution and utilization
- \* Synthesize and develop the idea of regional dimensions.

#### **Course Content:**

**Unit 1-** India: Physiographic Divisions, seasons, drainage, Soil and Natural vegetation  
Distribution of Population by Race, and Language of India.

**Unit 2-** Economic features of India: Mineral and Power Resources: Distribution and Utilization of Iron Ore, Coal, Petroleum, Gas;

**Unit 3-** Agricultural Production of Rice, Wheat; Industrial Corridors and Industrial Regions of India

**Unit 4** Regional Account of Jharkhand: Physiography, Drainage, Climate, natural vegetation, Population and tribes (Santhal, Oraon, Munda);

**Unit 5-** Economic features of Jharkhand: Agriculture, minerals and industry -iron and steel industry, silk; Tourism

#### **Reference Books:**

- \* Deshpande, C. D., (1992): India: A Regional Interpretation, ICSSR, New Delhi.
- \* Johnson, B. L. C., ed. (2001): Geographical Dictionary of India. Vision Books, New Delhi.
- \* Khullar, D.R. (2014): India: A Comprehensive Geography, Kalyani Publishers, New Delhi.
- \* Majid Husain (2009): Geography of India, Tata McGraw hill Education Private Ltd, New Delhi.
- \* Mandal, R. B. (ed.), (1990): Patterns of Regional Geography An International Per.. Vol. 3 Indian Perspective.
- \* Pathak, C. R. (2003): Spatial Structure and Processes of Development in India. Regional Science Ass., Kolkata.
- \* Sharma, T.C. (2013): Economic Geography of India. Rawat Publication, Jaipur.

- \* Singh R. L., (1971): India: A Regional Geography, National Geographical Society of India.
- \* Singh, Jagdish., (2003): India - A Comprehensive & Systematic Geography, Gyanodaya Praka, Gorakhpur.

## MINOR COURSE- MN 1B PR: MINOR PRACTICALS-1B PR - CARTOGRAPHIC TECHNIQUE

Marks: Pr (ESE: 3Hrs) = 25

Pass Marks: Pr (ESE) = 10

(Credits: Practicals-01) 30 Hours

### *Instruction to Question Setter for*

#### End Semester Examination (ESE):

There will be one Practical Examination of 3Hrs duration. Evaluation of Practical Examination may be as per the following guidelines:

|                           |            |
|---------------------------|------------|
| Experiment                | = 15 marks |
| Practical record notebook | = 05 marks |
| Viva-voce                 | = 05 marks |

### Course Objective:

The Learning objective of this course are as follows-

- \* To familiarise students about methods, tools and techniques of mapping
- \* To make student learn and apply thematic mapping techniques and preparation of maps

### **Learning Outcomes:**

After the completion of course, the students will have ability to:

1. Have sound knowledge regarding elements of maps, methods to draw maps.
2. Have proper utilization of maps for the planning and development.
3. Appreciate the preparation of various thematic maps with the application of various techniques.

### **Course Content:**

**Unit 1-** Thematic Mapping Techniques – Properties, Uses and Limitations; Areal Data -- Choropleth, Dot, Proportional Circles; Point Data – Isopleths.

**Unit 2-** Diagrammatic Data Presentation – Line, Bar and Circle; Cartographic Overlays – Point, Line and Areal Data. Thematic Maps – Preparation and interpretation

### **Reference Books**

- \* Cuff J. D. and Mattson M. T., 1982: Thematic Maps: Their Design and Production, Methuen Young Books
- \* Dent B. D., Torguson J. S., and Holder T. W., 2008: Cartography: Thematic Map Design (6th Edition), McGraw-Hill Higher Education
- \* Gupta K. K. and Tyagi V. C., 1992: Working with Maps, Survey of India, DST, New Delhi. Kraak M.-J. and Ormeling F., 2003: Cartography: Visualization of Geo-Spatial Data, Prentice- Hall.
- \* Mishra R. P. and Ramesh A., 1989: Fundamentals of Cartography, Concept, New Delhi. Sharma J. P., 2010: Prayogic Bhugol, Rastogi Publishers, Meerut.
- \* Singh R. L. and Singh R. P. B., 1999: Elements of Practical Geography, Kalyani Publishers.

## MINOR COURSE-1C (SEM-V)

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### MINOR COURSE- MN 1C: ENVIRONMENTAL GEOGRAPHY & SUSTAINABLE DEVELOPMENT

|                                                         |                                 |
|---------------------------------------------------------|---------------------------------|
| Marks: 15 (5 Attd. + 10 SIE: 1Hr) + 60 (ESE: 3Hrs) = 75 | Pass Marks: Th (SIE + ESE) = 30 |
|---------------------------------------------------------|---------------------------------|

(Credits: Theory-03) **45 Hours**

#### **Course Objective:**

The Learning objective of this course are as follows-

- \* To familiarise students about structure, function of ecosystem, environmental problems
- \* To make student learn about sustainable development,

#### **Course Learning Outcomes:**

After the completion of course, the students will have ability to:

- \* Appreciate the structure and functions of ecosystems with examples
- \* Understand the environmental problems and relevant management strategies
- \* Understand the sustainable development, good governance, national environmental policy

#### **Course Content:**

**Unit 1-** Environmental Geography: Concepts and Approaches; Ecosystem – Concept and Structure; Ecosystem Functions.

**Unit 2-** Environmental Problems and Management: Air Pollution; Solid and Liquid Waste; Biodiversity Loss.

**Unit 3-** Sustainable Resource Development: Definition, Components and Limitations.

**Unit 4-** The Millennium Development Goals: National Strategies and International Experiences.

**Unit 5-** Sustainable Development Policies and Programmes: The proposal for SDGs at Rio+20; SDGs; Principles of Good Governance; National Environmental Policy.

#### **Reference Books:**

- \* Anand, Subhash (2010) Solid Waste Management, Mittal Publication, New Delhi.
- \* Casper, J.K. (2010) Changing Ecosystems: Effects of Global Warming. Info base Pub. New York.
- \* Kumaraswamy K., Alagappa Moses A., and M. Vasanthy (2018) Glimpses of Environmental Sciences, Notion Press, Chennai.
- \* Miller, G.T. (2007) Living in the Environment: Principles, Connections, and Solutions, Brooks/Cole Cengage Learning, Belmont.
- \* Agyeman, Julian, Robert D. Bullard and Bob, Evans., (Eds.) (2003): Just Sustainabilities: Development in an Unequal World. London: Earthscan. (Introduction and conclusion.).

- \* Baker, Susan., (2006): Sustainable Development. Milton Park, Abingdon, Oxon; New York, N.Y.: Routledge.
- \* Lohman, Larry., (2003): Re-imagining the population debate, Corner House Briefing.

## MINOR COURSE- MN 1C PR: MINOR PRACTICALS-1C PR- STATISTICAL METHODS IN GEORAPHY

Marks: Pr (ESE: 3Hrs) = 25

Pass Marks: Pr (ESE) = 10

(Credits: Practicals-01) **30 Hours**

### *Instruction to Question Setter for*

### End Semester Examination (ESE):

*There will be one Practical Examination of 3Hrs duration. Evaluation of Practical Examination may be as per the following guidelines:*

|                                  |                   |
|----------------------------------|-------------------|
| <i>Experiment</i>                | <i>= 15 marks</i> |
| <i>Practical record notebook</i> | <i>= 05 marks</i> |
| <i>Viva-voce</i>                 | <i>= 05 marks</i> |

### Course Objective:

The Learning objective of this course are as follows-

- \* To explain the concept of quantitative information in Geographical study.
- \* To explain the importance and sources of data
- \* To familiarise students about methods of graphic data representations

### Course Learning Outcomes:

After the completion of course, the students will have ability to:

- \* Use statistical methods and techniques in geographical analysis
- \* Understand quantitative data, graphical data representation.
- \* Understand ways and sources of primary and secondary data.

### **Course Content:**

- ❖ **Unit 1-** Sources of Data- primary, secondary; Measures of central tendency- Mean, median and mode
- ❖ **Unit 2-** Graphic representation- histogram, Ogive, polygons

### **Reference Books:**

- ❖ Mahmood A., 1977: Statistical Methods in Geographical Studies, Concept. Pal S. K., 1998: Statistics for Geoscientists, Tata McGraw Hill, New Delhi.
- ❖ Sarkar, A. (2013) Quantitative geography: techniques and presentations. Orient Black Swan Private Ltd., New Delhi
- ❖ Silk J., 1979: Statistical Concepts in Geography, Allen and Unwin, London. Spiegel M. R.: Statistics, Schaum's Outline Series.
- ❖ Yeates M., 1974: An Introduction to Quantitative Analysis in Human Geography, McGraw Hill, New York. Shinha, Indira (2007) Sankhyiki bhugol. Discovery Publishing House, New Delhi.

## MINOR COURSE-1D (SEM-VII)

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### MINOR COURSE- MN 1D: CLIMATE CHANGE VULNERABILITY AND ADAPTATION

|                                                          |                                 |
|----------------------------------------------------------|---------------------------------|
| Marks: 15 (5 Attnd. + 10 SIE: 1Hr) + 60 (ESE: 3Hrs) = 75 | Pass Marks: Th (SIE + ESE) = 30 |
|----------------------------------------------------------|---------------------------------|

(Credits: Theory-03) **45 Hours**

#### **Course Objective:**

The Learning objective of this course are as follows-

To familiarise students about climate change, global warming

To make student learn about vulnerability, adaptation and mitigation to climate change

#### **Course Learning Outcomes:**

After the completion of course, the students will have ability to:

Understand the foundational concepts of climate change and its impacts.

Assess the human and environmental vulnerability to climate change.

Learn the various adaptation and mitigation for reducing the impacts of climate change and national action plan.

#### **Course Content:**

**Unit 1-** Climate Change: Understanding Climate Change; Greenhouse Gases and Global Warming; Global Climatic Assessment- IPCC

**Unit 2-** Climate Change and Vulnerability: Physical Vulnerability; Economic Vulnerability; Social Vulnerability

**Unit 3-** Impact of Climate Change: Agriculture and Water; Flora and Fauna; Human Health

**Unit 4-** Adaptation and Mitigation: Global Initiatives with Particular Reference to South Asia.

**Unit 5-** National Action Plan on Climate Change; Local Institutions (Urban Local Bodies, Panchayats)

#### **Reference Books:**

IPCC (2014): Climate Change 2014: Impacts, Adaptation, and Vulnerability. Part A: Global and Sectoral Aspects. Contribution of Working Group II to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.

OECD (2008): Climate Change Mitigation: "What do we do?" (Organisation and Economic Co-operation and Development).

Sen, Roy, S., and Singh, R.B., (2002): Climate Variability, Extreme Events and Agricultural Productivity in Mountain Regions, Oxford & IBH Pub., New Delhi.

Singh, M., Singh, R.B., and Hassan, M.I., (Eds.) (2014): Climate change and biodiversity, Proceedings of IGU Rohtak Conference, Volume 1. Advances in Geographical and Environmental Studies, Springer

Singh, R.B., Mal, Suraj, and Huggel, Christian (2018): Climate Change, Extreme Events and Disaster Risk Reduction, Springer, Switzerland, pages 309.

**MINOR COURSE- MN 1D PR:  
MINOR PRACTICALS-1D PR-  
INTRODUCTION OF REMOTE SENSING, GIS, GPS**

Marks: Pr (ESE: 3Hrs) = 25

Pass Marks: Pr (ESE) = 10

(Credits: Practicals-01) 30 Hours

***Instruction to Question Setter for***

***End Semester Examination (ESE):***

*There will be one Practical Examination of 3Hrs duration. Evaluation of Practical Examination may be as per the following guidelines:*

*Experiment = 15 marks*

*Practical record notebook = 05 marks*

*Viva-voce = 05 marks*

**Course Objective:**

The Learning objective of this course are as follows-

To familiarise students to use satellite remote sensing imagery, data interpretation

To make students learn application of GIS, GPS technology

**Course Learning Outcomes:**

After the completion of course, the students will have ability to:

Use and apply methods of remote sensing, GIS and GPS

Apply technology in solving many real time problems and issues

**Course Content:**

**Unit 1-** Remote sensing- Definition, types and its application; platform and sensor, TCC, FCC and satellite (Landsat and IRS) image interpretation, Land Use /Land Cover mapping

**Unit 2-** GIS: Definition, Components and its Application, Raster and Vector Data Structure, creation of vector data- point, line, polygon; Global Positioning System (GPS) – Principles and Uses, waypoint collection using handheld GPS or mobile phone (geotagging)

**References-**

1. Bhatta , B., (2008): *Remote Sensing and GIS*, Oxford University Press, New Delhi.
2. Campbell, J. B., (2007): *Introduction to Remote Sensing*, Guildford Press
3. Chauniyal, D., (2010): *Sudur Samvedana Avam Bhaugolik Suchna Pranali*, Sharda Pustak Bhawan, Allahabad.
4. Hord R.M.,(1989): *Digital Image Processing of Remotely Sensed Data*, Academic, New York.
5. Jensen, J. R., (2005): *Introductory Digital Image Processing: A Remote Sensing Perspective*, Pearson Prentice-Hall.
6. Jensen, J. R.,(2007): *Remote Sensing of the Environment: An Earth Resource Perspective*, Prentice-Hall Inc, New Jersey.
7. Joseph, G.,(2005): *Fundamentals of Remote Sensing*, United Press India.